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Inductive aexp : Type :=
| ANum (n : nat)
| APlus (a1 a2 : aexp)
| AMinus (a1 a2 : aexp)
| AMult (a1 a2 : aexp).

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Fixpoint aeval (a : aexp) : nat :=
match a with
| ANum n => n
| APlus a1 a2 => (aeval a1) + (aeval a2)
| AMinus a1 a2 => (aeval a1) - (aeval a2)
| AMult a1 a2 => (aeval a1) * (aeval a2)
end.

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What does the following expression evaluate to?

`aeval (APlus (ANum 3) (AMinus (ANum 4) (ANum 1)))`

$$\begin{aligned}
 & \text{aeval} \quad \begin{array}{c} + \\ / \quad \backslash \\ 3 \quad - \\ \quad / \quad \backslash \\ \quad 4 \quad 1 \end{array} = \underbrace{\text{aeval } 3}_3 + \text{aeval } \begin{array}{c} \overline{\quad} \\ / \quad \backslash \\ 4 \quad 1 \end{array} \\
 & \quad \quad \quad + \left(\underbrace{\text{aeval } 4}_4 - \underbrace{\text{aeval } 1}_1 \right) \\
 & \quad \quad \quad = 3 + (4 - 1) \\
 & \quad \quad \quad = 6
 \end{aligned}$$

$$\begin{array}{c}
\text{E_Skip} \frac{}{st \xrightarrow{\text{skip}} st} \quad \text{E_Asgn} \frac{a \Rightarrow_{st} n}{st \xrightarrow{x := a} st[x \mapsto n]} \quad \text{E_Seq} \frac{st \xrightarrow{c_1} st' \quad st' \xrightarrow{c_2} st''}{st \xrightarrow{c_1; c_2} st''}
\end{array}$$

$$\begin{array}{c}
\text{E_IfTrue} \frac{b \Rightarrow_{st} \text{true} \quad st \xrightarrow{c_1} st'}{st \xrightarrow{\text{if } b \text{ then } c_1 \text{ else } c_2 \text{ end}} st'} \quad \text{E_IfFalse} \frac{b \Rightarrow_{st} \text{false} \quad st \xrightarrow{c_2} st'}{st \xrightarrow{\text{if } b \text{ then } c_1 \text{ else } c_2 \text{ end}} st'}
\end{array}$$

$$\begin{array}{c}
\text{E_WhileTrue} \frac{b \Rightarrow_{st} \text{true} \quad st \xrightarrow{c} st' \quad st' \xrightarrow{\text{while } b \text{ do } c \text{ end}} st''}{st \xrightarrow{\text{while } b \text{ do } c \text{ end}} st''} \quad \text{E_WhileFalse} \frac{b \Rightarrow_{st} \text{false}}{st \xrightarrow{\text{while } b \text{ do } c \text{ end}} st}
\end{array}$$

$$\frac{4 \Rightarrow_{\{x \rightarrow 2\}} 4}{\text{E_Asgn} \quad \frac{2 := 4}{\{x \rightarrow 2\} \xrightarrow{} st'}} \quad \text{E_IfFalse}$$

$$\frac{2 \Rightarrow_{\{x\}} 2}{\{x\} \xrightarrow{x := 2} \{x \rightarrow 2\}} \quad \frac{x \leq 1 \Rightarrow_{\{x \rightarrow 2\}} \text{false}}{\{x \rightarrow 2\} \xrightarrow{\text{if } x \leq 1 \text{ then } y := 3 \text{ else } z := 4 \text{ end}} \{z \rightarrow 4; x \rightarrow 2\}}$$

$$\text{E_Seq} \frac{\{x\} \xrightarrow{x := 2} \{x \rightarrow 2\} \quad \{x \rightarrow 2\} \xrightarrow{\text{if } x \leq 1 \text{ then } y := 3 \text{ else } z := 4 \text{ end}} \{z \rightarrow 4; x \rightarrow 2\}}{\{ \} \xrightarrow{x := 2; \text{ if } x \leq 1 \text{ then } y := 3 \text{ else } z := 4 \text{ end}} \{z \rightarrow 4; x \rightarrow 2\}}$$

$$\begin{array}{c}
\mathbf{E_Skip} \quad \frac{}{st \xrightarrow{\text{skip}} st} \qquad \mathbf{E_Asgn} \quad \frac{a \Rightarrow_{\text{st}} n}{st \xrightarrow{x := a} st[x \mapsto n]} \qquad \mathbf{E_Seq} \quad \frac{st \xrightarrow{c_1} st' \quad st' \xrightarrow{c_2} st''}{st \xrightarrow{c_1; c_2} st''}
\end{array}$$

$$\begin{array}{c}
\mathbf{E_IfTrue} \quad \frac{b \Rightarrow_{\text{st}} \text{true} \quad st \xrightarrow{c_1} st'}{st \xrightarrow{\text{if } b \text{ then } c_1 \text{ else } c_2 \text{ end}} st'} \qquad \mathbf{E_IfFalse} \quad \frac{b \Rightarrow_{\text{st}} \text{false} \quad st \xrightarrow{c_2} st'}{st \xrightarrow{\text{if } b \text{ then } c_1 \text{ else } c_2 \text{ end}} st'}
\end{array}$$

$$\begin{array}{c}
\mathbf{E_WhileTrue} \quad \frac{b \Rightarrow_{\text{st}} \text{true} \quad st \xrightarrow{c} st' \quad st' \xrightarrow{\text{while } b \text{ do } c \text{ end}} st''}{st \xrightarrow{\text{while } b \text{ do } c \text{ end}} st''} \qquad \mathbf{E_WhileFalse} \quad \frac{b \Rightarrow_{\text{st}} \text{false}}{st \xrightarrow{\text{while } b \text{ do } c \text{ end}} st}
\end{array}$$